BPL_STEM_AIR_Perfusion - test

Here we show simulations of stem cell cultivation in an aerated hollow fiber reactor. The reactor volume is kept constant and cells recycled, thus the setup is similar to perfusion cultivation.

The model combines rudimentary cell growth and metabolism combined with times series data of of the metabolic rates: qNmax, qLc, and qO2, marked with red in the comprehensive plot.

Ref Greuel et al: "Online measurement of oxygen enables continuous noninvasive evalation of human-induced pluripotent stem cell (hiPSC) culture in a perfused 3D hollow-fober bioreactor", Biotech. Bioeng., 2019.

In [1]:
```
# Import the application model and context variables
from BPL_STEM_AIR_Perfusion import*
```

```
Windows - run FMU pre-compiled JModelica 2.14
```

In [2]:
```
# Impprt the general FMU_explore functions ana adapt them to the application
from fmu_explore_pyfmi import fmu_explore

Dummy = fmu_explore(model, parValue, parLocation, parCheck,  fmu_model, fmu_proc
                    MSL_usage, MSL_version, BPL_version, \
                    opts_std, simulationTime, timeDiscreteStates, stateValue, \
                    diagrams, ax, lines)
par = Dummy.par
init = Dummy.init
disp = Dummy.disp
simu = Dummy.simu
show = Dummy.show
resetPen = Dummy.resetPen
process_diagram = Dummy.process_diagram
describe_general = Dummy.describe_general
describe_parts = Dummy.describe_parts
describe_MSL = Dummy.describe_MSL
FMU_explore_info = Dummy.FMU_explore_info
system_info = Dummy.system_info
SDG = Dummy.SDG
```

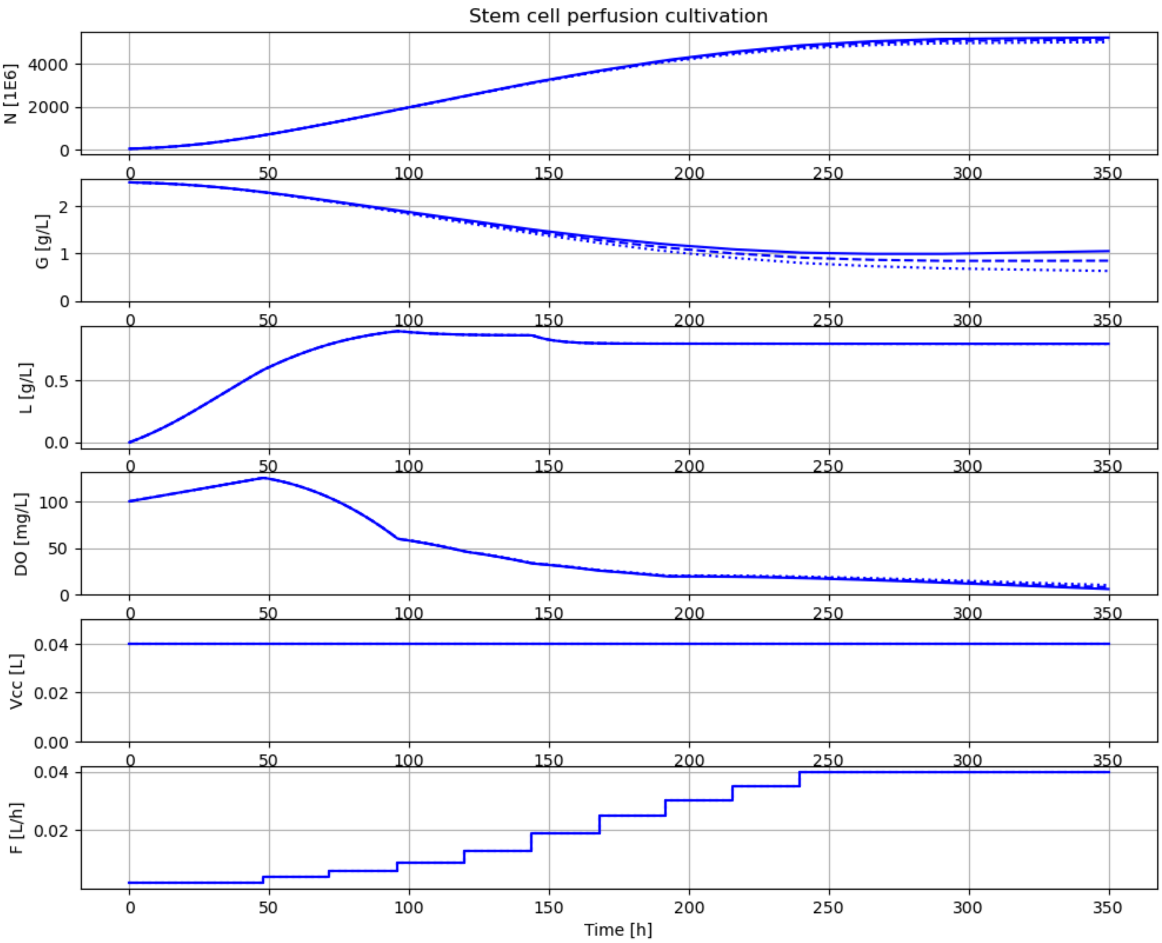
In [3]:
```
run -i BPL_STEM_AIR_Perfusion_explore.py
```

```
Model for the process has been setup. Key commands:
 - par()       - change of parameters and initial values
 - init()      - change initial values only
 - simu()      - simulate and plot
 - newplot()   - make a new plot
 - show()      - show plot from previous simulation
 - disp()      - display parameters and initial values from the last simulation
 - describe()  - describe culture, broth, parameters, variables with values/units

Note that both disp() and describe() takes values from the last simulation
and the command process_diagram() brings up the main configuration

Brief information about a command by help(), eg help(simu)
Key system information is listed with the command system_info()
```

In [4]:
```
plt.rcParams['figure.figsize'] = [30/2.54, 24/2.54]
```

In [5]:
```
#process_diagram()
```

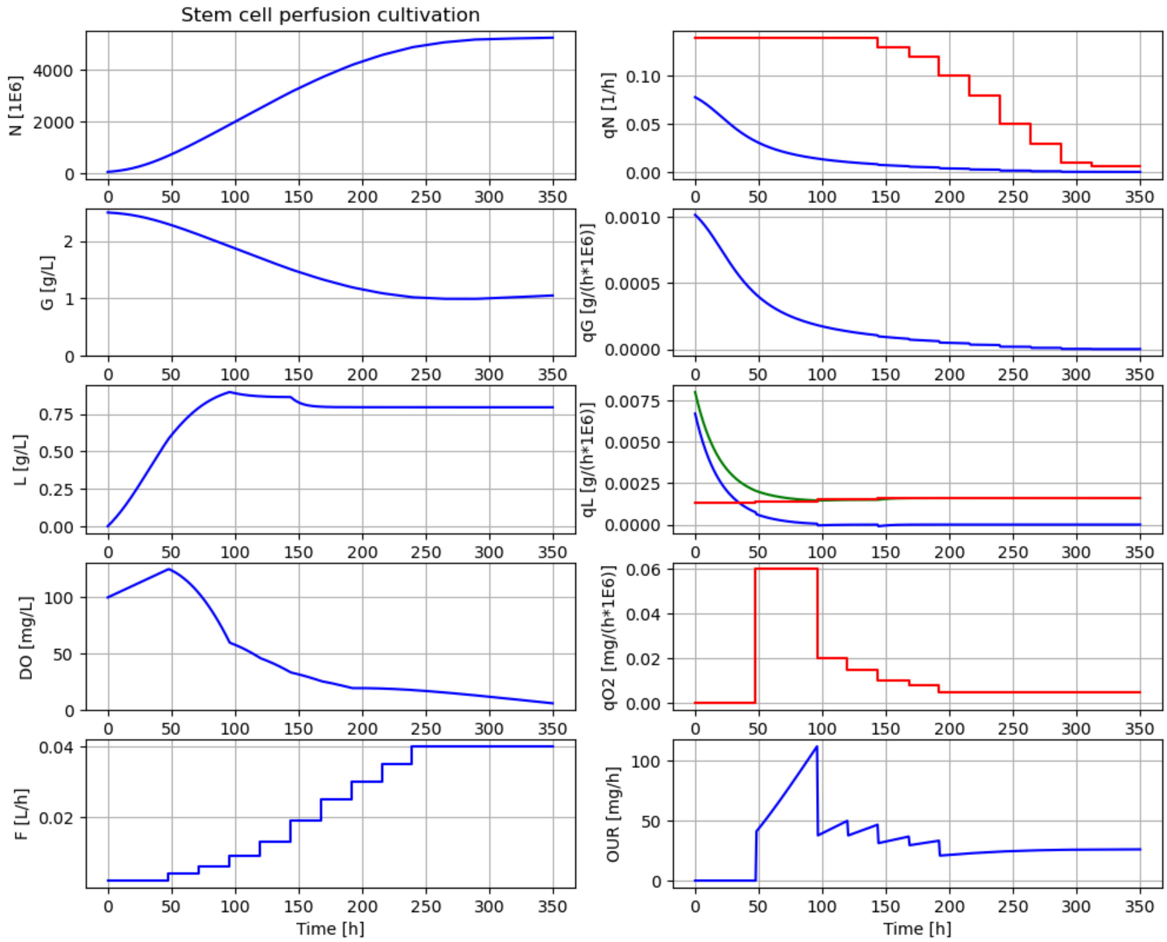
In [6]:
```
# Process parameters to mimic stem cell example
# - study impact of different values maintenatnce qm

newplot(plotType='Basic')
for value in [1.0e-6, 10.0e-6, 20e-6]:
    par(qm=value)
    simu(350)
```



In [7]:
```
# Process parameters to mimic stem cell example
# . comprehensvie plot including metabolic rates

par(qm=1e-6)
```

```
newplot(plotType='Comprehensive')
simu(350)
```

Out[7]: <pyfmi.fmi_algorithm_drivers.FMIResult at 0x2cd34709a30>



In [8]:
```
describe('culture')
```

Reactor culture human-induced pluripotent stem cells - hiPSCs

In [9]:
```
describe('Vcc')
```

Volume of the reactor : 0.04 [L]

In [10]:
```
describe('N_start')
```

Number of cells at start : 50.0 [1E6]

In [11]:
```
describe('N')
```

Number of cells : 5223.752 [1E6]

In [12]:
```
describe('G_start')
```

Glucose conc at start : 2.5 [g/L]

In [13]:
```
describe('G_in')
```

Glucose feed conc : 2.5 [g/L]

In [14]:
```
describe('L')
```

Lactate conc : 0.796 [g/L]

In [15]:
```
describe('DO')
```

```
Dissolved oxygen conc : 6.045 [ mg/L ]
```

Appendix

In [16]:
```
system_info()
```

```
System information
 -OS: Windows
 -Python: 3.12.11
 -Scipy: not installed in the notebook
 -PyFMI: 2.21.0
 -FMU by: JModelica.org
 -FMI: 2.0
 -Type: FMUModelCS2
 -Name: BPL_STEM.Reactor
 -Generated: 2025-07-22T09:58:37
 -MSL: 4.1.0
 -Description: BPL - not used
 -Interaction: FMU-explore version 1.1.5
```